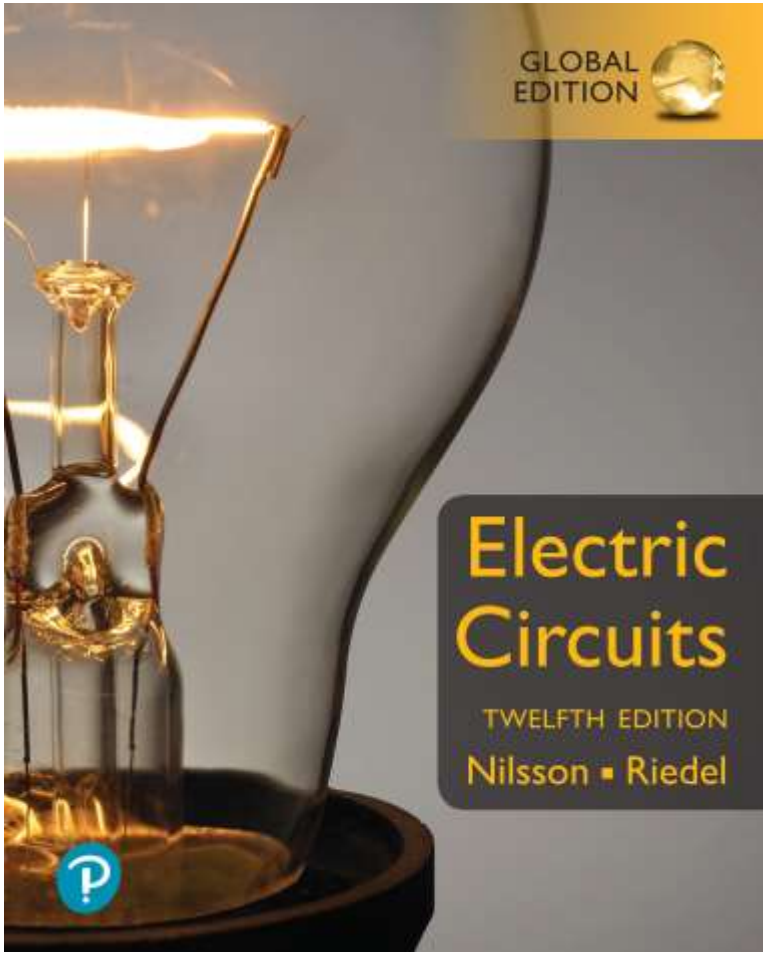
Electric Circuits

Twelfth Edition, Global Edition



Chapter 2 – Part 2

Circuit Elements

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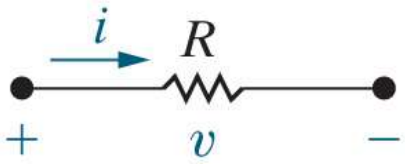
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Chapter 2 – Circuit Elements (1 of 22)

Resistance

- A measure of the capacity of a material to impede the flow of current.
- The property of resistance is modeled by a resistor.
- A resistor is also a circuit element that has a specified resistance associated with it.

Resistor



v is the voltage in volts [V]

i is the current in amps [A]

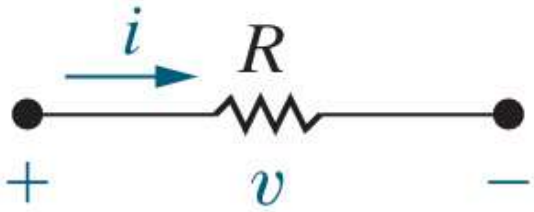
R is the resistance in ohms [Ω]

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Chapter 2 – Circuit Elements (2 of 22)

Resistors



Ohm's Law describes the relationship between voltage and current in a resistor:

$$v = Ri$$

v is the voltage in volts [V]

i is the current in amps [A]

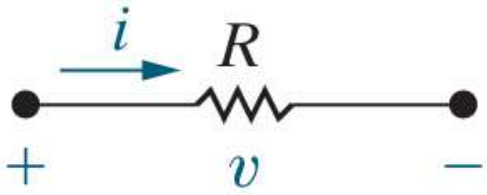
R is the resistance in ohms [Ω]

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Chapter 2 – Circuit Elements (3 of 22)

Not quite – the voltage in Ohm's law is a signed quantity.



$\boxed{v = \pm Ri}$ Ohm's Law

How do you determine which sign to use? Use the passive sign convention!

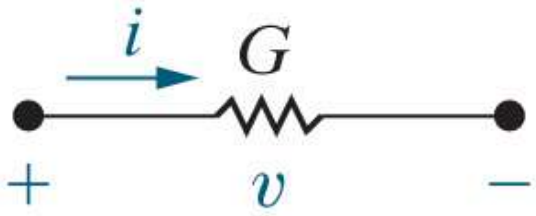
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Chapter 2 – Circuit Elements (4 of 22)

More on resistors:

The inverse of resistance is **conductance.**



$$G = \frac{1}{R}; \qquad i = \pm Gv$$

Conductance has the units Siemens, abbreviated “S”.

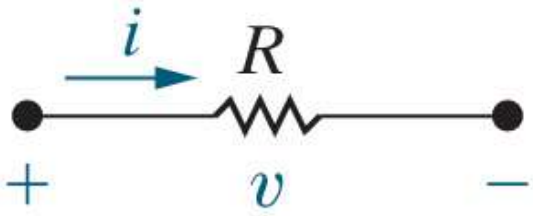
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Chapter 2 – Circuit Elements (5 of 22)

More on resistors

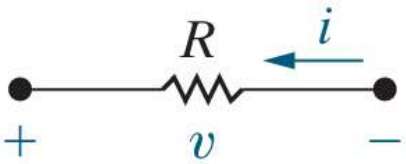
- Calculate the power in a resistor:



$$p = +vi = (Ri)i$$

$$= Ri^2 > 0!$$

The power associated with a resistor is always positive. Therefore, resistors always absorb power!



Find the power for this resistor.

$$\boxed{p = -vi = -(-Ri)i = Ri^2 > 0!}$$

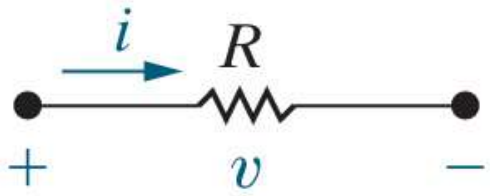
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Chapter 2 – Circuit Elements (6 of 22)

More on resistors

- Calculate the power in a resistor using voltage:



$$p = +vi = v\frac{v}{R}$$

$$= \frac{v^2}{R} > 0!$$

The power associated with a resistor is still positive.

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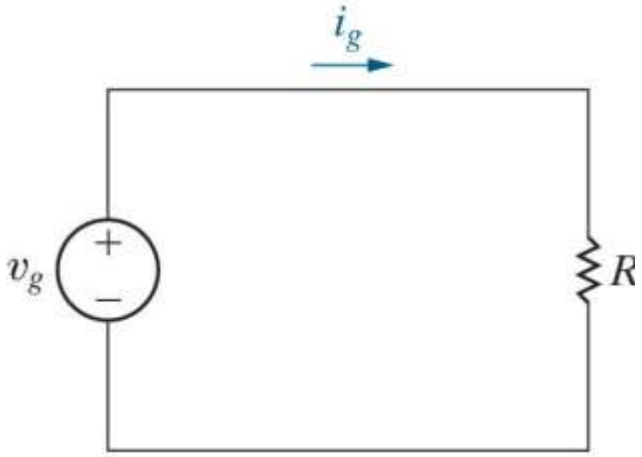
Chapter 2 – Circuit Elements (7 of 22)

Example:

- Consider the circuit shown here.

(a) If $v_g = 1\,\text{kV}$ and $i_g = 5\,\text{mA}$, find

R and the power absorbed by the resistor.



$$R = +\frac{v_g}{i_g} = \frac{1000\text{ V}}{5\text{ mA}} = \frac{1000}{0.005} = 200{,}000\Omega = 200\text{ k}\Omega$$

$$p_R = Ri_g^2 = (200{,}000)(0.005)^2 = 5\text{ W}$$

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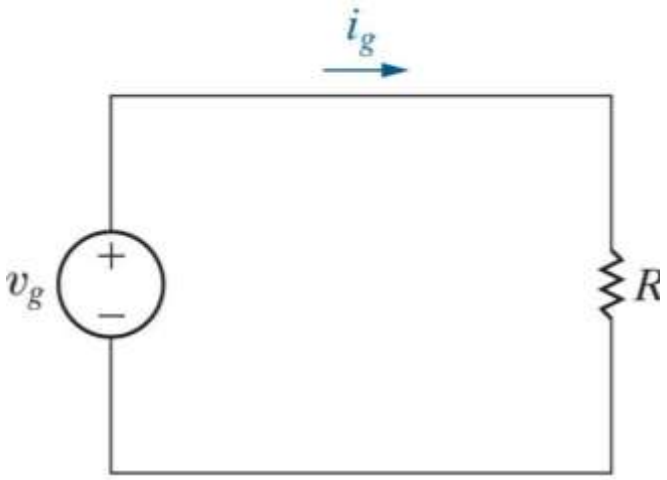
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Chapter 2 – Circuit Elements (8 of 22)

Example, continued

- Consider the circuit shown here.

(b) If $i_g = 75$ mA and $p_g = -3$ W, find v_g, R and the power absorbed by the resistor.



$$p_g = -v_g i_g \quad \Rightarrow \quad v_g = -\frac{p_g}{i_g} = -\frac{(-3)}{0.075} = 40 \text{ V}$$

$$\mathrm{R} = \frac{\mathrm{v_g}}{\mathrm{i_g}} = \frac{40}{0.075} = 533.33\Omega$$

$$p_R = Ri_g^2 = (533.33)(0.075)^2 = 3 \text{ W}$$

OR... $p_R = -p_g = -(-3) = 3$ W!

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Chapter 2 – Circuit Elements (9 of 22)

We just calculated the power of the resistor to be 3 W. Is the resistor supplying or absorbing power?

- ✗ A. Supplying
- ✓ B. Absorbing

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Chapter 2 – Circuit Elements (10 of 22)

Suppose we are told that the voltage drop across a $50\ \Omega$ resistor is 10 V. What is the power associated with the resistor?

- ✗ A. 5000 W
- ✗ B. 500 W
- ✓ C. 2 W
- ✗ D. I need to know the voltage polarity first

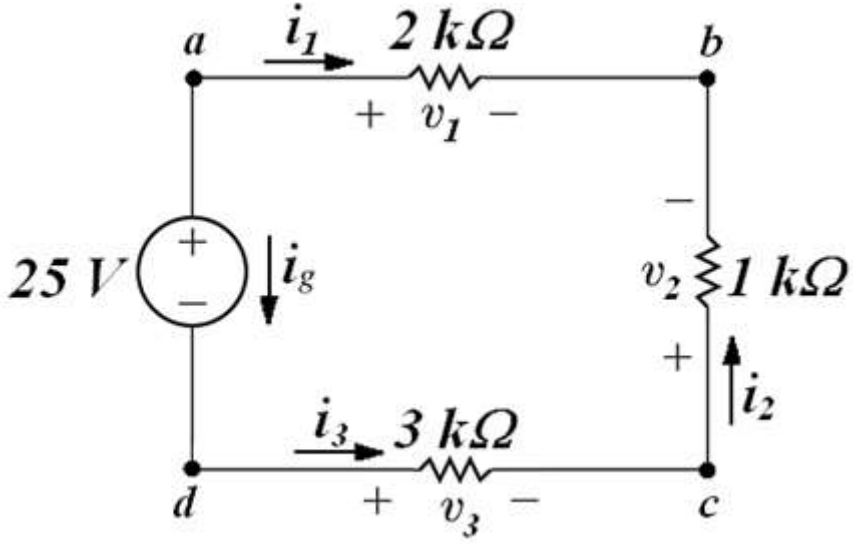
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Chapter 2 – Circuit Elements (11 of 22)

Kirchhoff's Laws

- Consider the circuit shown below. We wish to "solve" this circuit. To solve a circuit, we must find all of the unknown voltages and currents.



How many unknowns are in this circuit? Seven – three unknown voltages and four unknown currents.

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Chapter 2 – Circuit Elements (12 of 22)

To solve for 3 unknown voltages and 4 unknown currents, how many independent equations do we need?

- ✗ A. 4
- ✓ B. 7
- ✗ C. 8
- ✗ D. We need more information

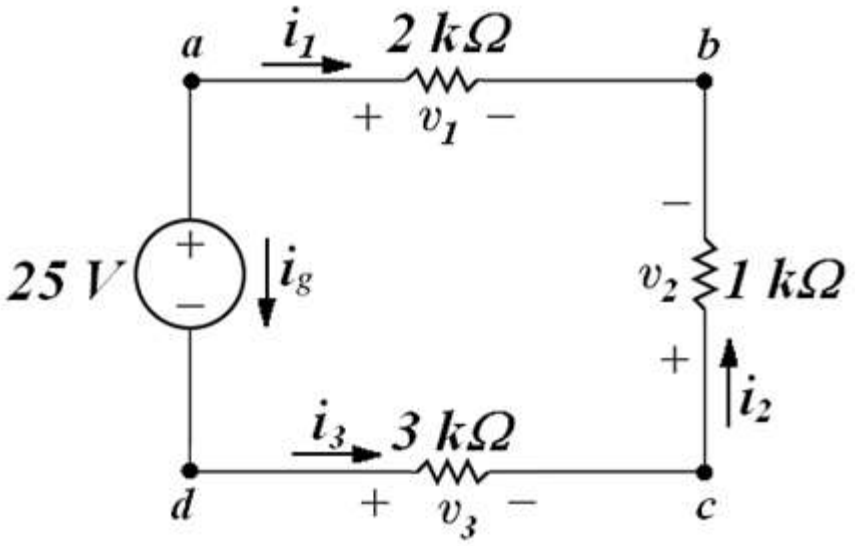
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Chapter 2 – Circuit Elements (13 of 22)

Kirchhoff's Laws (continued)

- The only equations we know how to write at this point come from Ohm's law. Let's write them.



$$v_1 = 2000 i_1$$
$$v_2 = 1000 i_2$$
$$v_3 = 3000 i_3$$

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Chapter 2 – Circuit Elements (14 of 22)

Kirchhoff's Laws (continued)

- We need more equations – they will come from Kirchhoff's laws.

Kirchhoff's Current Law (KCL)

- The algebraic sum of all the currents at any node in a circuit equals zero.

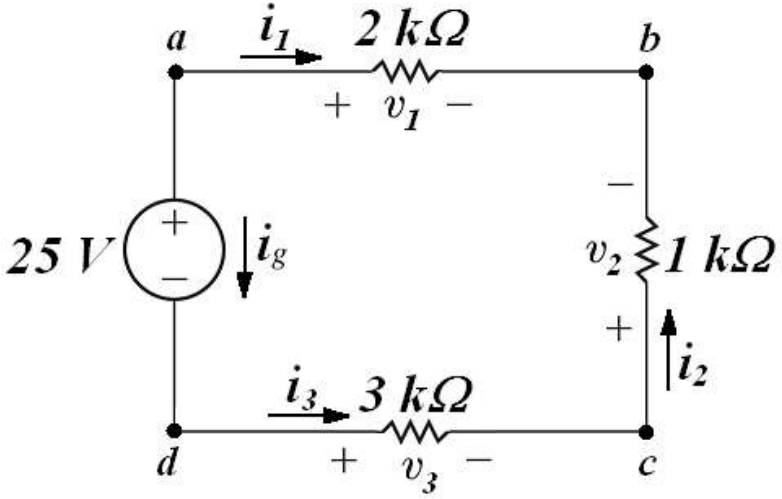
Note – a **node** is a point where two or more circuit elements are connected.

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Chapter 2 – Circuit Elements (15 of 22)

How many nodes are in this circuit?



- ✔ A. 4
- ✘ B. 3
- ✘ C. 2
- ✘ D. None of the above

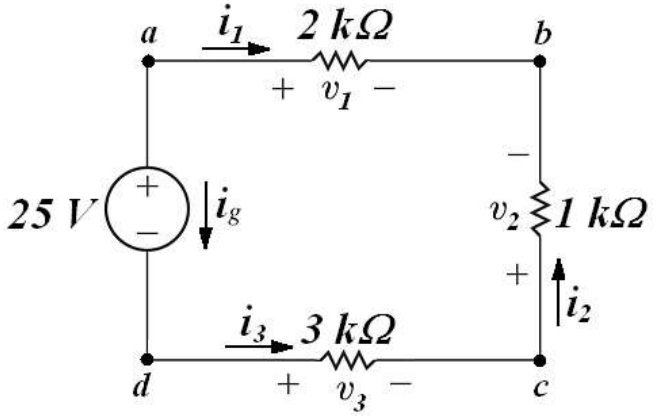
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Chapter 2 – Circuit Elements (16 of 22)

Kirchhoff's Current Law (KCL)

- Let's write the KCL equations at each node. Are they all independent?



Note that only 3 of the 4 are independent – the 4th equation can be derived from the other 3.

a (leaving):	$i_g + i_1 = 0$
b (entering):	$i_1 + i_2 = 0$
c (enter = leave):	$i_2 = i_3$
d (leaving):	$-i_g + i_3 = 0$

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Chapter 2 – Circuit Elements (17 of 22)

If a circuit has 6 nodes, how many independent KCL equations can you write?

- ✗ A. 7
- ✗ B. 6
- ✓ C. 5
- ✗ D. None of the above

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Chapter 2 – Circuit Elements (18 of 22)

Kirchhoff's Laws (continued)

- We have 3 equations from Ohm's law and 3 (independent) KCL equations – we need one more equation, because we have 7 unknowns. That equation will come from Kirchhoff's Voltage Law

Kirchhoff's Voltage Law (KVL)

- The algebraic sum of all the voltages around any closed path in a circuit equals zero.

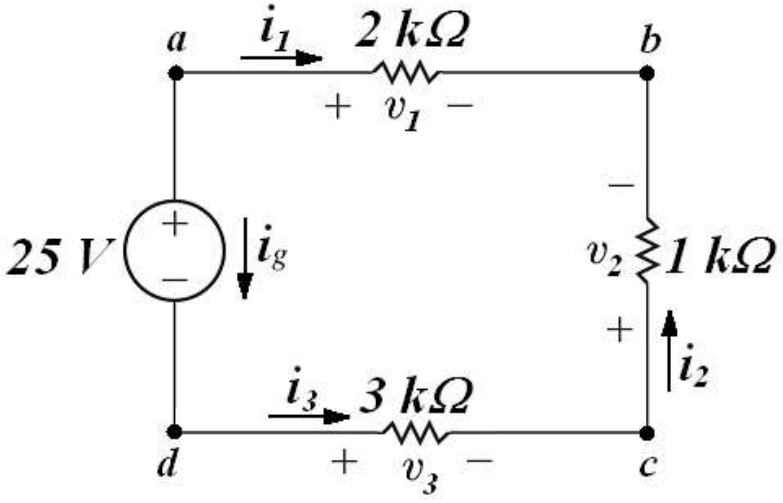
Note – a closed path starts at any node, travels through selected circuit elements, and returns back to the starting node without including any intermediate nodes more than once.

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Chapter 2 – Circuit Elements (19 of 22)

Kirchhoff's Voltage Law (KVL)

- Let's write the KVL equation around the closed path formed by this circuit.



Start at node a and go clockwise:

$$+v_1 - v_2 - v_3 - 25 = 0$$

Start at node d and go counter-clockwise:

$$+v_3 + v_2 - v_1 + 25 = 0$$

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Chapter 2 – Circuit Elements (20 of 22)

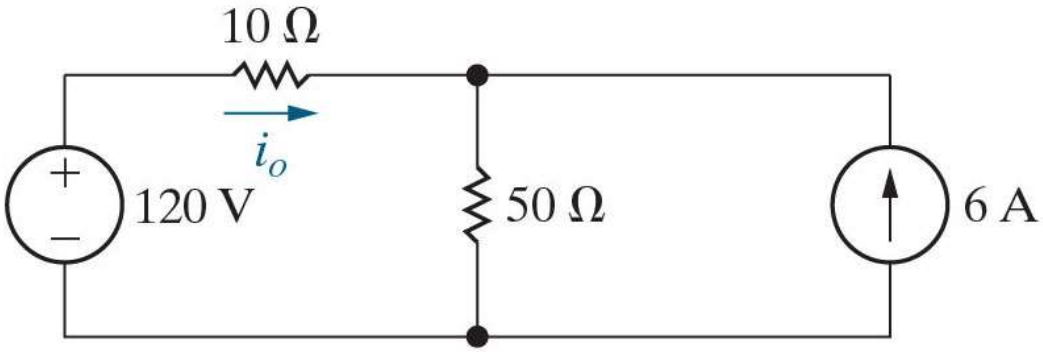
Kirchhoff's Laws (continued)

- Now we have 7 equations and 7 unknowns, so we can plug the 7 equations into a calculator and solve them to solve the circuit.
- Analysis of circuits with resistors and constant sources is based on the use of Ohm's Law, KCL and KVL. However, we will develop some useful analysis techniques that will NOT require us to solve 7 equations for the simple example circuit we've been studying. Actually, we can solve this circuit with just 1 equation!

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Chapter 2 – Circuit Elements (21 of 22)

How many voltages and currents are in this circuit?



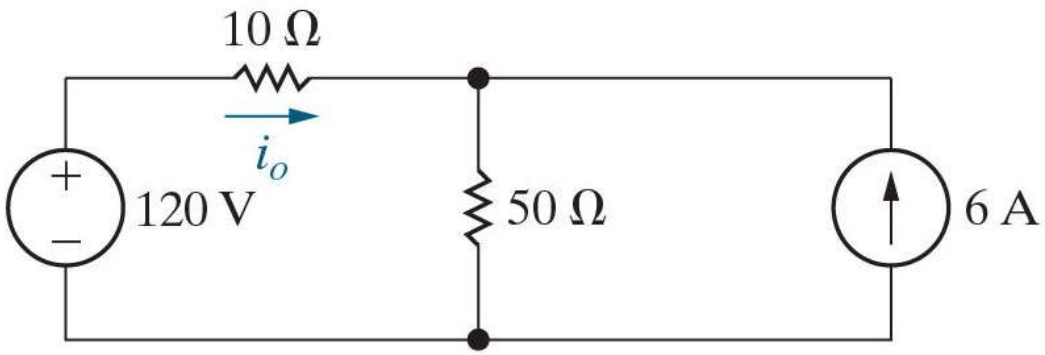
- ✗ A. 4 voltages; 4 currents
- ✗ B. 4 voltages; 3 currents
- ✗ C. 3 voltages; 4 currents
- ✓ D. 3 voltages; 3 currents

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Chapter 2 – Circuit Elements (22 of 22)

How many Ohm's Law, KCL, and KVL equations will we need to solve this circuit?



✔ A. 2 OL, 2 KCL, 2 KVL

✘ B. 2 OL, 1 KCL, 3 KVL

✘ C. 2 OL, 3 KCL, 1 KVL

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